Life Expectancy Changes Due to Kidney Donation Across Race, Sex, and Age

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Nicholas S. Selby[1] and X[2]

Abstract
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Keywords
keyword1, keyword2, keyword3

Background

Design

Model overview

I constructed a Markov state-transition model to compare lifetime outcomes for two hypothetical cohorts of otherwise identical individuals: one cohort that donates a kidney and one that does not. Both cohorts enter the model at the age of donation in full health, and the simulation advances in annual cycles until age 110. The analysis was conducted from the perspective of the individual donor rather than the health-care system; outcomes are reported as undiscounted remaining life-years, consistent with the standard approach for life-expectancy analyses.[1] The base-case age at donation is 40 years; subgroup analyses span ages 25–55. I used a Monte Carlo simulation with 5×10^6 individuals per arm and cross-validated against a deterministic analytic cohort implementation.

Health states

The model contains seven mutually exclusive health states (Figure 1): *Healthy*, *ESRD onset*, *Dialysis*, *Standard waitlist*, *Priority waitlist*, *Post-transplant*, and *Dead* (absorbing). At ESRD onset, the cohort branches: a fraction is listed preemptively (bypassing dialysis entirely) and the remainder initiates haemodialysis. The donor arm routes ESRD survivors to the *Priority waitlist*, reflecting the allocation priority afforded to prior living donors (PLD) under KAS250; the non-donor arm routes to the *Standard waitlist*. Graft failure returns patients to the *Dialysis* state, bypassing the ESRD-onset branch.

ESRD risk after donation

Fifteen-year cumulative ESRD incidence was taken from Muzaale et al.[2]: 30.8 per 10,000 (95% CI 24.3–38.5) for donors and 3.9 per 10,000 (95% CI 0.8–8.9) for age-, sex-, and health-matched controls (8-fold relative risk). Race-specific donor rates were 74.7/10,000 (Black) and 22.7/10,000 (White).[2] The White non-donor baseline was taken from Grams et al.[3] (sex-averaged 0.05%/15 yr), because zero ESRD events were recorded in White matched controls in Muzaale et al.

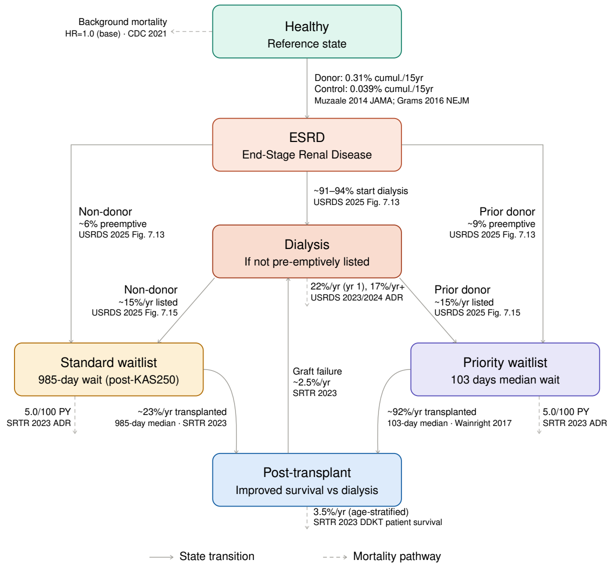

Figure 1. The Markov model for kidney donation.

I modelled ESRD hazard with a Weibull distribution (shape $k = 1.5$), calibrated via a competing-risk bisection algorithm to reproduce the 15-year cumulative incidence while accounting for background-mortality attrition in the at-risk pool. The accelerating hazard ($k > 1$) is supported by the Massie et al. 20-year cumulative incidence of 34 per 10,000 (vs. $\approx$21/10,000 predicted under a constant-hazard model).[4] I drew within-donor subgroup hazard ratios from Massie et al.: 2.96 (95% CI 2.25–3.89) for Black race, 1.88 (95% CI 1.50–2.35) for male sex, and 1.40 per decade (non-Black donors).[4]

[1]Renewvia Solar Ethiopia PLC
[2]SAGE Publications Ltd, UK

Corresponding author:
Nicholas S. Selby, Renewvia Solar Ethiopia PLC House No. 174/175, Saya Building, 5th Floor Bole sub-city, Woreda 03 Addis Ababa, Ethiopia
Email: nicholas.selby@renewvia.com

Dialysis survival and waitlist listing

First-year haemodialysis mortality was 22% and subsequent-year mortality 17% (≈42% five-year survival), consistent with USRDS estimates.[5]

At ESRD onset, a fraction of patients is listed preemptively before dialysis initiation, avoiding dialysis mortality entirely. I drew preemptive listing probabilities from USRDS 2025 ADR Figure 7.13: 5.8% overall for the non-donor arm and 9.4% for the donor arm (age 18–44 cohort, reflecting the younger and healthier donor-like population).[6]

Among patients on dialysis, the conditional annual probability of joining the waitlist was set to 15%/year (base case), derived from USRDS 2025 ADR Figure 7.17.[6] Approximately 25% of 18–44 year-old ESRD patients are listed or transplanted within one year, of whom ≈9.4% were preemptively listed, yielding an estimated post-dialysis listing rate of ≈15.6%/year for the donor-like cohort (conservative base case: 15%/year; one-way sensitivity range 5–30%/year).

Transplant waiting time and waitlist outcomes

Transplant probability while listed was modelled as a time-homogeneous exponential process parameterised by the median waiting time, a standard approximation for waiting-time analyses.[7] For the standard waitlist, I used the post-KAS250 national median of 985 days;[8] for the PLD priority waitlist, I used the post-KAS median of 102.6 days was used.[9] Waitlist mortality was 5.0 deaths per 100 patient-years (SRTR 2023 ADR Figure KI 24) and competing removal 6.81%/year (SRTR 2023 ADR Figure KI 22).[10]

Post-transplant survival

I derived post-transplant annual mortality from age-stratified 5-year Kaplan–Meier estimates for DDKT recipients in the SRTR 2023 ADR (Figure KI 70, 2016–2018 cohort): 0.9%/year (ages 18–34), 1.8%/year (35–49), 3.9%/year (50–64), and 6.9%/year (≥ 65).[10] The base case uses DDKT survival because prior living donors who develop ESRD predominantly receive transplants via the deceased-donor priority pathway rather than living-donor transplantation, consistent with USRDS data showing that living-donor transplants account for a small minority of transplants in this population.[11,12] A one-way sensitivity analysis substitutes LDKT patient survival (SRTR 2023 ADR Figure KI 76: 0.4%, 0.8%, 1.7%, and 3.9%/year by age band) to bound the effect of this assumption. Annual graft failure was 2.5%/year post-year 1;[10] graft failure returned patients to the Dialysis state.

Background mortality and donor all-cause survival

I took background mortality in the Healthy state from the 2021 US Life Tables (CDC, National Vital Statistics Reports Vol. 72 No. 12),[13] with sex-specific tables used in sex-stratified analyses. The base-case donor all-cause mortality hazard ratio versus matched non-donors is 1.0, consistent with US-based studies[2,14] and the Grams et al. 2018 meta-analysis (pooled HR 0.984, 95% CI 0.743–1.302; 9 studies, $n = 118{,}426$ donors).[15] I also examined an HR of 1.30 from Mjøen et al.[16], derived from a Norwegian cohort in which 80% of donors were first-degree relatives of the recipient, as the pessimistic sensitivity scenario.

Subgroup analyses

Age at donation I repeated the analyses at ages 25, 35, 40, 45, and 55 years at donation. Donor ESRD rates were scaled by the Massie 2017 age HR (1.40/decade, non-Black donors);[4] I held non-donor baselines at the Muzaale overall matched-control rate.

Race I took race-specific donor ESRD rates (Black 74.7/10,000; White 22.7/10,000) and race-specific non-donor baselines (Black sex-averaged 0.195%/15 yr; White sex-averaged 0.05%/15 yr) from Muzaale et al.[2] and Grams et al., respectively.[3] Race-specific waitlist mortality and post-transplant survival were from SRTR 2023 Figures KI 25 and KI 71.[10]

Sex I derived sex-specific donor ESRD rates were derived by applying the Massie 2017 male-sex HR (1.88)[4] to the overall donor rate, weighted by the observed donor sex mix (60% female; SRTR 2023 ADR Figure KI 7).[10] For non-donor baselines, we used the Grams 2016 White sex-specific rates (female 0.04%/15 yr; male 0.06%/15 yr) rather than applying the within-donor sex HR to the control arm, because Massie 2017 hazard ratios describe variation within the donor cohort and are not appropriate for scaling a different reference population.[3] I applied this sourcing rule, Muzaale matched-control rates for overall analyses and Grams direct rates wherever sex or race is stratified, consistently across all simulations.

Probabilistic sensitivity analysis

I conducted a PSA with 200 parameter draws. Sampled quantities and their distributions were:

- ESRD 15-year cumulative risks: beta distributions parameterised from the Muzaale 2014 published 95% CIs;
- Donor all-cause mortality HR: log-normal, $\mu = -0.016$, $\sigma = 0.143$, derived from the Grams 2018 meta-analysis CI;
- Weibull shape k: log-normal, $\sigma = 0.13$, representing structural uncertainty absent published CIs for this parameter; and
- Waitlist listing probability: beta, fractional SE = 40%, consistent with the 0.05–0.30 plausible range

Registry-derived parameters (from SRTR or USRDS) have negligible sampling error relative to structural uncertainty and were fixed at base-case values; scenario variation in these parameters is captured by the one-way sensitivity analysis. I report the mean ΔLE, 95% credible interval, and the probability that donation is life-expectancy–neutral or beneficial.

One-way sensitivity analysis

Each parameter was varied individually while others were held at base-case values. Key scenarios included: no priority access (PLD wait set to the standard 985-day median), donor all-cause mortality HR = 1.30 (Mjøen pessimistic estimate), ESRD relative risk $\times 11$ (Mjøen upper-bound ESRD HR), PLD wait 50 days (optimistic), post-transplant survival at LDKT quality (SRTR Figure KI 76), and dialysis mortality $\pm 50\%$. Results are reported in Section .

Calibration

The model's transplant probability was validated against the SRTR 2023 3-year waitlist outcome distribution (Figure KI 22, 2019–2021 listing cohort)[10]. This comparison is non-circular because the transplant probability was derived from the median waiting time, not from the 3-year outcome distribution. The model predicted 44.3% transplanted at three years versus 44.3% observed (DDKT + LDKT combined; $\Delta = 0\%$). The waitlist removal rate was back-calculated via a competing-risk bisection algorithm to reproduce the SRTR-observed 19.1% removed at three years (12.6%/year); the naïve formula $1 - (1 - \mathrm{CIF})^{1/3}$, which ignores competing depletion by transplantation and death, was not used. The model over-predicted three-year waitlist mortality (10.0% vs. 6.8% observed, $\Delta = +47\%$), likely reflecting the application of a 2023 cross-sectional mortality rate (confirmed 5.0/100 patient-years from the raw SRTR data file) to a 2019–2021 listing cohort; this bias is symmetric across both arms and conservative (slightly overstates the cost of the waitlist period for both donors and non-donors equally). No independent cohort exists tracing the complete pathway from living donation through ESRD onset to transplantation; these checks represent the available face-validity evidence for the model.

Voucher benefit and ESRD-conditional priority benefit

Two supplementary analytic-cohort Markov models were built on the same state structure, transition probabilities, and parameter sources as the base-case donor model, in order to decompose the population-level donor ΔLE result into its constituent mechanisms.

The first model quantifies the life-expectancy effect of a kidney-donation "voucher": a mechanism by which a living donor designates a healthy, non-donor family member to receive PLD priority-waitlist access should that family member ever develop ESRD. Two arms of healthy non-donors were simulated, both carrying non-donor baseline ESRD risk and a background all-cause mortality HR of 1.0; the sole difference between arms was waitlist priority upon ESRD onset (voucher arm: PLD median wait 102.6 days[9]; control arm: standard median wait 985 days[8]). Consistent with the sourcing rule used for the base-case preemptive-listing parameters, the voucher arm was assigned the informed/monitored preemptive-listing probability (9.4%) and the control arm the general-population probability (5.8%)[6]. All other transition probabilities (dialysis mortality, per-cycle waitlist listing, waitlist mortality and removal, post-transplant survival, graft failure) were identical to the base-case model[10]. The analysis was run at the base-case age of 40, with an age sweep (25–60 years) and an ESRD-risk subgroup analysis using the Muzaale overall and Grams race-specific non-donor CIFs[2,3]. One-way sensitivity analyses varied the voucher and standard median wait times, the per-cycle listing probability, the preemptive-listing probability, and post-transplant graft quality (DDKT vs. LDKT).

Because the voucher (and, by extension, the donor PLD-priority) benefit is diluted at the population level by the large fraction of the cohort who never develop ESRD, a second model conditioned entirely on ESRD onset: all individuals entered the model already diagnosed with ESRD, in order to isolate the life-expectancy benefit of priority waitlist access for the sub-population who actually reach the waitlist. Entry to dialysis versus preemptive listing, dialysis and waitlist mortality, waitlist removal, post-transplant survival, and graft failure used the same age-stratified, all-cause parameters as the base-case model; no separate Healthy state or background-mortality term was included, since dialysis- and waitlist-state mortality rates are all-cause rates already measured in ESRD populations. Two arms were compared, differing only in waitlist priority (priority vs. standard median wait, as above). The base case used an ESRD-onset age of 60, with an age sweep (40–75 years at ESRD onset) and one-way sensitivity analyses on wait times, per-cycle listing probability, preemptive-listing probability, post-transplant graft quality, and dialysis mortality ($\pm 20\%$).

Both supplementary models used a deterministic analytic-cohort implementation (as in the base-case cross-validation) with 10^6 individuals per arm.

Analytic cycle length and discounting

Annual (one-year) cycles were used throughout. Life-years accumulated with an end-of-cycle convention; the resulting systematic underestimate of approximately 0.5 years applies equally to both arms and cancels in ΔLE. Future life-years were not discounted, consistent with the convention for pure life-expectancy analyses[1].

Software and reproducibility

All analyses were implemented in Python 3.10 (NumPy, pandas, matplotlib). The simulation, parameter assembly pipeline, sensitivity analyses, and calibration checks are fully reproducible from the public code repository at https://github.com/rupumped/kidney-donor-le.

Results

Base-case life expectancy

In the base-case analysis (age 40 at donation, overall population parameters), kidney donation was associated with a life-expectancy cost of $\Delta\mathrm{LE} = -37.0$ days (donor 38.15 yr vs. non-donor 38.16 yr remaining life expectancy; analytic cohort). The Monte Carlo simulation (5×10^6 individuals per arm) confirmed this estimate ($\Delta\mathrm{LE} = -38.3$ days). Across all base-case and subgroup scenarios, donation was associated with a net life-expectancy cost to the donor; no scenario identified donation as neutral or beneficial to the individual.

Subgroup analyses

Age at donation The life-expectancy cost decreased monotonically with older age at donation, from -52.8 days at age 25 to -16.3 days at age 55 (Table 1). This gradient

Table 1. Life-expectancy results by subgroup. LE Donor and LE Non-donor are remaining life-years from the age at donation. ΔLE is the donor minus non-donor difference; negative values indicate a life-expectancy cost of donation. All results are from the analytic cohort implementation (deterministic); the base-case Monte Carlo (5×10^6/arm) is shown for comparison.

Subgroup	LE Donor (yr)	LE Non-donor (yr)	ΔLE (days)
Base case			
Age 40 (analytic cohort)	38.15	38.16	−37.0
Age 40 (Monte Carlo)	38.05	38.15	−38.3
Age at donation			
Age 25	51.64	51.78	−52.8
Age 35	42.52	42.64	−43.0
Age 40	38.05	38.16	−37.0
Age 45	33.67	33.75	−30.3
Age 55	25.27	25.31	−16.3
Race (age 40)			
White	38.08	38.15	−25.1
Overall	38.04	38.15	−37.0
Black	37.87	38.09	−73.4
Sex (age 40)			
Female	40.45	40.48	−29.9
Overall	38.06	38.17	−37.0
Male	35.97	36.09	−43.6

reflects the shorter remaining lifetime over which excess ESRD risk accumulates: younger donors have more life-years at risk and thus more opportunity for the elevated ESRD hazard to materialize.

Racial disparity Race was the strongest subgroup modifier of the life-expectancy cost (Figure 2). Black donors experienced a cost of -73.4 days, nearly three times the -25.1 days estimated for White donors at age 40. The disparity widened sharply with younger age at donation: a Black donor at age 25 loses -199 days of life expectancy compared with -35 days for a White donor of the same age. This disparity traces directly to post-donation ESRD incidence: Black donors have a 15-year cumulative incidence of 74.7 per 10,000 versus 22.7 per 10,000 in White donors[2], a 3.3-fold difference that is further compounded by a higher background ESRD rate in age- and health-matched Black non-donors[3]. These results indicate that existing tools for counselling prospective donors substantially underestimate the life-expectancy impact for Black individuals.

Sex Male donors faced a larger life-expectancy cost (-43.6 days) than female donors (-29.9 days) at the base-case age of 40, consistent with the male-sex within-donor ESRD hazard ratio of 1.88 reported by Massie et al.[4]. Sex and race effects compound multiplicatively: a Black male donor at age 40 loses -92 days, roughly double the -44 days for an overall male and six times the -14 days for a White female donor of the same age (Figure 2). At age 25 the compounding is more extreme still, with Black male donors losing -249 days versus -29 days for White female donors.

Probabilistic sensitivity analysis

The PSA (200 parameter draws) produced a mean ΔLE of $+29.5$ days (95% credible interval $-1{,}085.5$ to $+1{,}068.0$ days; P[donation beneficial] $= 48\%$; Figure 3). The wide credible interval and near-50% probability of benefit are driven almost entirely by uncertainty in the donor all-cause mortality hazard ratio (see Section), which produces a bimodal distribution: a tight cluster near -37 days when the HR is near its point estimate of 1.0, and a second cluster near $-1{,}068$ days when the HR is drawn from the upper tail of the log-normal prior. Conditional on the HR remaining at 1.0 — the value supported by the US-based literature and the Grams et al. meta-analysis[15] — the model result is robustly and consistently negative.

One-way sensitivity analysis

The donor all-cause mortality hazard ratio dominated all other parameters by an order of magnitude (Table 2, Figure 4). Setting HR $= 1.30$[16] increased the life-expectancy cost to $-1{,}067.9$ days, a swing of more than 1,000 days from base case. No other parameter produced a swing exceeding 32 days. Dialysis mortality ($\pm 50\%$) shifted ΔLE by at most ± 12 days. Variation in the standard waitlist wait time across the plausible range (730–1,760 days) altered ΔLE by less than 0.2 days.

The priority waitlist had similarly negligible influence: removing PLD priority access entirely — setting the donor waitlist to the standard 985-day median — worsened ΔLE by only 1.9 days relative to base case. Halving the priority wait to 50 days recovered only 0.1 days. This insensitivity arises because fewer than 0.3% of donors develop ESRD within 15 years and fewer still ever reach the transplant waitlist; the priority pathway is simply too rarely invoked to move a population-level outcome meaningfully.

Decomposition of the priority-waitlist and voucher benefits

Donor priority benefit Although KAS250 affords prior living donors (PLD) priority access to the deceased-donor waitlist[9], the resultant life-expectancy benefit to the donor is negligible at the population level. The one-way analysis confirms that full removal of priority access costs the donor only 1.9 additional days, and that the observed priority benefit — diluted across the large fraction of donors who never develop ESRD — contributes less than 2 days to the overall donor ΔLE.

Voucher benefit The life-expectancy benefit of a kidney-donation voucher to a healthy, non-donor family member

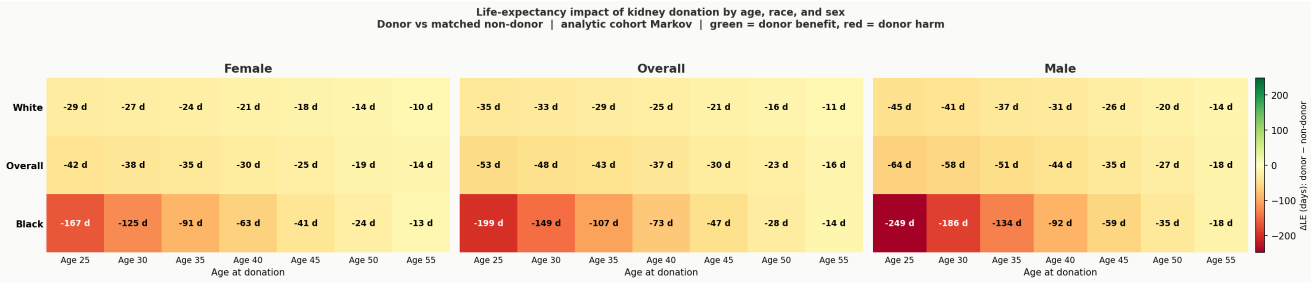

Figure 2. Life-expectancy impact of kidney donation by age, race, and sex. Each cell shows ΔLE (donor minus non-donor, days) for the indicated sex panel, race row, and age-at-donation column (analytic cohort Markov). All values are negative across every combination examined. Race and sex effects compound: a Black male donor at age 25 loses -249 days, compared with -10 days for a White female donor at age 55. Colour scale: red = larger donor harm, yellow = smaller harm.

Table 2. One-way sensitivity analysis: parameter values tested and the resulting change in life expectancy (donor arm minus non-donor arm), age 40 at donation, all other parameters held at base-case values. Negative ΔLE values indicate a net life-expectancy cost of donation relative to the matched non-donor arm under the value tested; positive values indicate a donor-arm benefit. The "$8\times$ controls (base case)" row reproduces the overall base-case result reported in the main results table (ΔLE $= -37.0$ d).

Parameter (value tested)	ΔLE (days)	vs. base (days)
ESRD relative risk, donor vs. non-donor (base case: $8\times$, 30.8/10,000)		
$2\times$ controls	-5.1	$+31.9$
$4\times$ controls	-15.9	$+21.1$
$8\times$ controls (base case)	-37.0	—
$11\times$ controls (Mjøen upper bound)	-53.7	-16.7
Standard waitlist median wait (base case: 985 d)		
24 months (730 d)	-37.0	$+0.0$
58 months (1760 d, pre-KAS250)	-36.8	$+0.2$
Priority (PLD) waitlist median wait (base case: 102.6 d)		
50 d (optimistic)	-36.9	$+0.1$
200 d	-37.2	-0.2
No priority access (985 d)	-38.9	-1.9
Dialysis annual mortality (base case: 17%)		
-50% (8.5%)	-24.6	$+12.4$
$+50\%$ (25.5%)	-43.9	-6.9
Donor all-cause mortality HR (base case: 1.0)		
HR = 1.30 (Mjøen pessimistic)	-1067.9	-1030.9
Post-transplant survival (base case: DDKT quality)		
LDKT quality (SRTR Figure KI 76)	-22.7	$+14.3$

peaked at $+0.8$ days for a recipient designated at age 25, falling to $+0.3$ days at age 40 and $+0.1$ days at age 60 (Figure 5). These small figures reflect the same dilution mechanism: conditional on developing ESRD and reaching the priority waitlist, priority access adds a clinically substantial $+605$ days of life expectancy for a patient whose ESRD onset occurs at age 40, and $+209$ days for onset at age 60 (Figure 6). However, since only approximately 0.04% of non-donors develop ESRD within 15 years, this large conditional benefit compresses to well under one day at the population level. Voucher benefit was modestly higher for Black non-donor family members ($+1.7$ days at age 40), owing to their approximately five-fold higher baseline ESRD incidence[3].

Donor life-expectancy cost in context

The mean life-expectancy cost to the individual donor ($\approx$ 37 days) must be weighed against the benefit conferred on the recipient and, through the voucher mechanism, on designated family members who may one day develop ESRD. Our ESRD-conditional model demonstrates that priority waitlist access alone is worth more than 200 days to a patient who has already developed ESRD; the full benefit of receiving a functioning kidney rather than remaining on long-term haemodialysis is larger still. Viewed in this frame, donation represents a transfer of life expectancy from donor to recipient on highly favourable terms: the donor foregoes approximately five weeks of expected life in exchange for providing the recipient with the prospect of years of additional survival.

Conclusions

Acknowledgements

Acknowledgements

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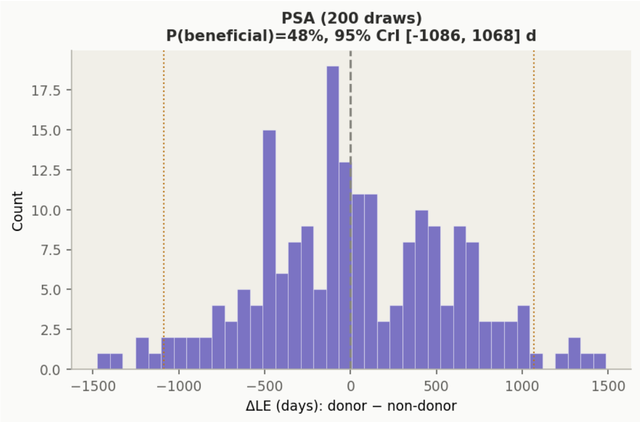

Figure 3. Probabilistic sensitivity analysis (PSA), 200 parameter draws. Histogram of ΔLE (donor minus non-donor, days). The dashed vertical line marks zero; orange dotted lines mark the 95% credible interval bounds. The distribution is bimodal: a cluster near the base-case value of -37 days (donor all-cause mortality HR ≈ 1.0) and a heavy left tail driven by draws in which the HR exceeds 1.0. The near-50% probability of benefit ($P = 48\%$) reflects this structural uncertainty in the HR rather than genuine equipoise in the base-case scenario.

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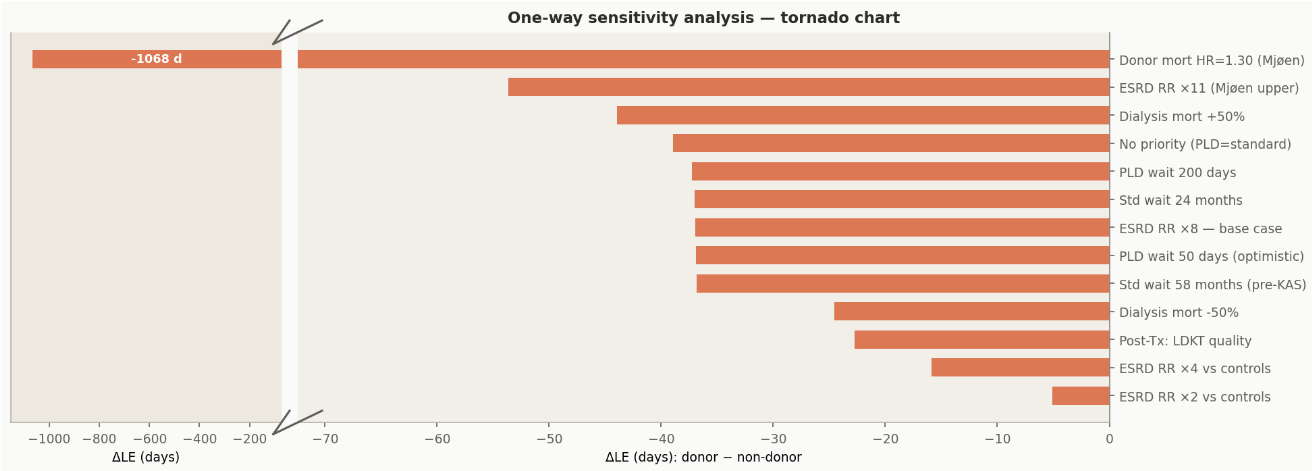

Figure 4. One-way sensitivity analysis — tornado chart. Each bar shows the ΔLE resulting from testing the indicated parameter value (see Table 2 for exact values). The donor all-cause mortality HR (top bar, truncated at left) produces a $> 1{,}000$-day shift that dwarfs every other parameter. The right panel uses a finer scale to show that all other parameters produce swings of ≤ 32 days. Priority waitlist parameters ("No priority" and "PLD wait" rows) are barely distinguishable from the base case, confirming the negligible population-level effect of transplant priority.

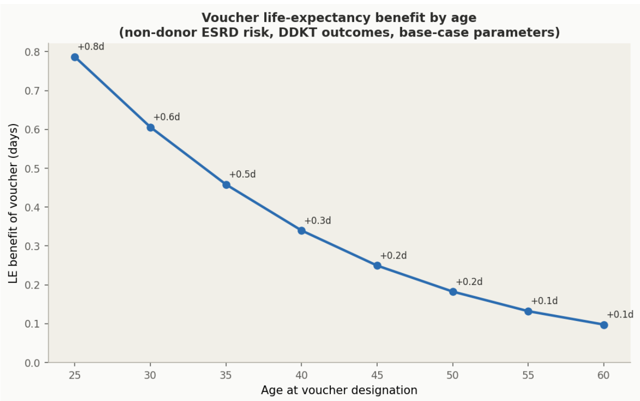

Figure 5. Voucher life-expectancy benefit by age at designation. ΔLE (voucher holder minus unvouched non-donor, days) for a healthy non-donor family member carrying a kidney-donation voucher, as a function of the age at which the voucher is designated. Both arms carry non-donor ESRD risk. The benefit declines with age because older individuals have less remaining lifetime over which the voucher's priority access could be exercised. Even at the maximum (age 25, $+0.8$ days), the benefit is small at the population level due to the low baseline ESRD incidence among non-donors.

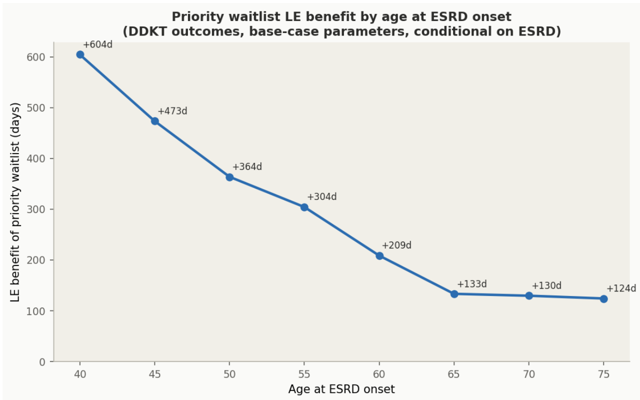

Figure 6. Priority waitlist LE benefit conditional on ESRD onset, by age. ΔLE (priority minus standard waitlist, days) for individuals who have *already developed* ESRD, as a function of age at ESRD onset. Conditional on ESRD, priority access is highly valuable: $+605$ days at age 40 onset, $+209$ days at age 60. This large conditional benefit is diluted to ≤ 0.8 days at the population level (Figure 5) by the low ESRD incidence in non-donors ($\approx 0.04\%$/15 yr) and in donors ($\approx 0.3\%$/15 yr).